

Q1 4. Buggy Code in Stack Push Operation

1 mark

```
#include <stdio.h>
#define MAX 5
int stack[MAX];
int top = -1;

void push(int data) {
    if (top == MAX - 1)
        printf("Stack Overflow");
    else
        stack[top++] = data;
}

int main() {
    push(10);
    push(20);

    printf("%d", stack[top]);
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #define MAX 5
3
4 int stack[MAX];
5 int top = -1;
6
7 void push(int data)
8 {
9     if (top == MAX - 1)
10        printf("Stack Overflow");
11    else
12        stack[++top] = data;
13 }
14
15 int main()
16 {
17     push(10);
18     push(20);
19
20     printf("%d", stack[top]);
21
22     return 0;
23 }
```

✓ Submitted**Test Input****Output**

20

🕒 Pending manual review

Q2 Buggy Code in Stack POP Operation

1 mark

```
#include <stdio.h>
#define MAX 5
int stack[MAX];
int top = 2;
int pop() {
    if (top == -1)
        return -1;
    return stack[++top];
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include<stdio.h>
2 #define MAX 5
3 int stack[MAX]={10,20,30};
4 int top=2;
5 int pop()
6 {
7     if(top==--1)
8     {
9         printf("underflow");
10        return -1;
11    }
12    return stack[top--];
13 }
14 int main()
15 {
16     printf("%d",pop());
17     return 0;
18 }
```

✓ Submitted

Test Input

stdin input (optional)

Output

30

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Q3 Buggy Code in underflow operation

1 mark

```
int pop() {
    if(top == 0) {
        printf("Underflow");
        return -1;
    }
    return stack[top--];
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #define MAX 5
3
4 int stack[MAX] = {10, 20, 30};
5 int top = 2;
6
7 int pop()
8 {
9     if(top == -1)
10    {
11        printf("Underflow\n");
12        return -1;
13    }
14    return stack[top--];
15 }
16
17 int main()
18 {
19     int item = pop();
20
21     if(item != -1)
22         printf("Popped element: %d\n", item);
23
24     return 0;
25 }
```

✓ Submitted

Test Input

stdin input (optional)

Output

Popped element: 30

⌚ Pending manual review